

Concentrated N-dimensional AMM with Polar Coordinates in Rust

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SUMMARY: We expand on the recent development of n-dimensional automated market makers for stablecoins by showing a way to build concentrated liquidity positions with ticks in polar coordinates in Rust, including the featured ability to skew said concentrated liquidity and highlight the risks of stacking too many stablecoin pools and how to reduce said risk. We also introduce multimodal liquidity distributions.

Keywords: Automated market maker, Concentrated liquidity, Concentrated circular market maker, Liquidity Provision, Liquidity fingerprint, Multimodal liquidity fingerprint, Polar coordinates, Rust, Superellipse, Ticks.

1. INTRODUCTION

There exist two schools of thought for liquidity concentration in Automated Market Makers (AMM). One approach involves the specification of a custom curve with parameters controlling the shape of the liquidity distribution [1] such as Curve [2] and Euler-swap [3]. The second approach consists in giving the Liquidity Provider (LP) the freedom to select a discrete price range where liquidity can be concentrated such as Uniswap v3 [4] and Orbital AMM [5]. Herein we focus on an n-dimensional AMM for stablecoins where we:

- (1) Explain the Orbswap invariant.
- (2) Demonstrate swap function in polar coordinates.
- (3) Concentrate ticks in polar coordinates.
- (4) Show limitations of Orbital symmetric ticks and propose ability to adjust liquidity with ellipse.
- (5) Highlight the risk of multi-dimensional stablecoin pools and show a path for mitigation.
- (6) Introduce multi-modal liquidity distributions.

2. Orbswap Invariant

Uniswap v2 introduced a continuous uniform liquidity distribution [1] allowing for the trading of a token in the price range of zero to infinity. Yet prices are not infinite nor bound by the zero barrier, to address part of this problem Uniswap v3 was introduced to allow for the sharp concentration of liquidity based on an LP's preferred viewpoint on the range of price movement between zero and infinity [4]. To achieve a softer concentration of liquidity we can adjust the Uniswap v3 AMM invariant equation below by allowing the curve to fold in on itself. A side effect of it is the ability to access negative prices as pointed out on ethresear.ch [6] which can be disabled for our stablecoin use case.

$$(x - k)^2 + (y - k)^2 = k^2 \quad (1)$$

where k is an offset parameter to pin liquidity. The liquidity fingerprint of this Concentrated Circular Market Maker (CCMM) is derived in Appendix B and can be extended to create Orbswap - a Concentrated Super-Elliptical Market Maker (CSEMM) to further widen/narrow and skew the liquidity allocation with the invariant becoming:

$$\left| \frac{x}{\alpha} - 1 \right|^{u(\alpha)} + \left| \frac{y}{\beta} - 1 \right|^{u(\beta)} = 1 \quad (2)$$

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where $u(x) = \frac{\ln(2)}{\ln(\frac{x}{x-1})}$. Parameters α and β allow for the extension of the liquidity in the left and right tail. By setting $\alpha = \beta$ and decreasing their values towards 2 one can approach the liquidity concentration of $x + y = k$, by widening α and β towards infinity one increases the tails of the distribution until it converges to the Logarithmic Market Scoring Rule (LMSR) invariant [1]. At $\alpha = \beta = 2 + \sqrt{2}$ we retrieve the CCMM from equation 1 and is the version that we have launched on Arbitrum with a six dimensional pool outlined on our [github](#) [7].

2.1. Polar Swap Function

Unlike traditional AMM swap functions which operate in the Cartesian plane by moving along a curve, we can rotate using [polar coordinates](#) [8] from a focal point with distance L which stands for liquidity. This rotation involves a particular step where the quantity of one stablecoin one wishes to insert Δ_x is exchanged for the quantity of a second stablecoin Δ_y while passing through a trigonometric conversion and is outlined in Rust in Appendix 3.3.

2.2. Ticks in Polar Coordinates

The additional step of trigonometric conversion for Orbswap can be used to construct ticks [4] in the polar coordinate system where each tick can represent a particular granularity level. For the sake of simplicity, if our pool initiation consists of n -tokens all priced initially at \$1.00, then the polar tick sits at 45 degrees and has the ability to range from 0 to 90 degrees. As the swap function passes through a particular tick discretized into degrees, it can adjust parameter L to change the price impact.

2.3. Tick Limitations with Circles

The Orbital AMM paper [5] provides symmetric ticks, but liquidity can be demonstrated to be skewed in stablecoin pools [Figure 1], making it capital inefficient to not skew liquidity. Since under the Orbswap superelliptical skewed format no-closed-form polar solution is possible, [we can still shift it with just an elliptical form approach](#) [9] where a and b determine the shift intensity.

$$\left(\frac{x}{a} - L\right)^2 + \left(\frac{y}{b} - L\right)^2 = L^2 \quad (3)$$

2.4. Risks of n-dimensional pools

In the event of a permanent stablecoin depeg we witness significant divergence loss for the LP and an accumulation of the unwanted depegged stablecoin. For assets with n -dimensional pools, the first $n > 2$ stablecoin example being Curve's Tricrypto pool (USDC, USDT, DAI) one can historically recall the event of Silicon Valley Bank in 2023 [10] where a

bankrun forced USDC to \$0.85, resulting in a serious run on the pool and a flight to USDT. The LP of this $n = 3$ pool, in the event of a total depeg would have suffered significant divergence loss had the peg not returned. Extending this concept to n stablecoin pools towards infinity poses a significant problem, considering it is building in fragility of assets that are inherently fat tailed [Figure 2].

To counter this Kessler-like syndrome [11] of adding more stablecoins to a pool, we suggest not exceeding such high dimensions and stacking stablecoin orbits that are related together and, if one wishes to expand to $n > 3$ orbits, and include stablecoins with a higher risk of depegging, then LPs should consider hedging their position with depeg insurance or other options products. Additional safeguards should be researched for handling depeg events. For example, a system which removes a depegged stablecoin from the allowed list of tokens in the Orb automatically if it reaches a certain price below \$1.00 while re-balancing the prices of the other unaffected assets.

3. Appendix

3.1. Orbswap Liquidity Fingerprint

The liquidity fingerprint can be viewed as the second derivative with respect to the square root of the price [1]. In traditional tick space t we get the following distribution of liquidity:

$$L(t) = \pm \frac{2ke^{\frac{3t}{2}}}{(1 + e^{2t})^{\frac{3}{2}}}. \quad (4)$$

The value function of an LP payoff is given by the Legendre transform with the Greeks for Delta= Δ_c , Gamma= Γ_c , and Theta= Θ_c being available in desmos: <https://www.desmos.com/calculator/wy74cucie0> [12].

3.2. Cartesian Swap Functions

By solving for y in equation (1) for the CCMM we get:

$$y = -\sqrt{2kx - x^2} + k \quad (5)$$

Adding Δ_x and Δ_y into our swap function to get the quantity of y for quantity x :

$$\Delta_y = -\sqrt{2k(x - \Delta_x) - (x - \Delta_x)^2} + k - y \quad (6)$$

The CSEMM swap function becomes:

$$y = -\beta \left(\left(1 - \left| \frac{x}{\alpha} - 1 \right|^{\frac{\ln(2)}{\ln(\frac{\alpha}{\alpha-1})}} \right)^{\frac{\ln(\frac{\beta}{\beta-1})}{\ln(2)}} - 1 \right) \quad (7)$$

Adding Δ_x and Δ_y into our swap function

$$\Delta_y = -\beta \left(\left(1 - \left| \frac{x + \Delta_x}{\alpha} - 1 \right|^{u(\alpha)} \right)^{\frac{1}{u(\beta)}} - 1 \right) - y \quad (8)$$

For Δ_x due to the asymmetric nature of the superellipse our swap function order for $u(x)$ is reversed:

$$\Delta_x = -\beta \left(\left(1 - \left| \frac{y + \Delta_y}{\alpha} - 1 \right|^{u(\beta)} \right)^{\frac{1}{u(\alpha)}} - 1 \right) - x \quad (9)$$

3.3. Rust Arbitrum Stylus Math

Rust WASM smart contracts from Arbitrum Stylus can help make Solidity contracts more expressive. For example, math in Solidity can be limited. In contrast, Rust has more math function support with its standard library, and has a large Rust cargo collection of math libraries for even more complex math operations. Some of these Rust crate libraries can even be used where the Rust standard library might need to be disabled for compiling WASM binaries.

Note 1: it is recommended to used fixed point math libraries, since floating point calculations can be non deterministic based on the hardware the math is computed on.

Note 2: the Rust math function below uses decimal values that aren't scaled. For EVM production deployments, these values should be scaled up in a similar way to how Solidity library prb-math uses WAD values to represent decimal values since the EVM currently only supports integer values.

Rust math function related to the swap function with Line referring to equations outlined in this [desmos model](#) [8]:

```
fn get_delta_x_function() -> f64 {
    let l : f64 = 10.0;           // Desmos Line 1
    let y_in : f64 = get_y_in();  // Desmos Line 9 // 6.07106781187 test value

    let radians_45_degrees = std::f64::consts::FRAC_PI_4;
    let radians_135_degrees : f64 = 3.0 * radians_45_degrees;
    let cos_expression = radians_135_degrees.cos();
    let l_cos_expression = l * cos_expression;
    let ratio = y_in / l;
    let ratio_squared = f64::powf(ratio, 2.0);
    let radicand = 1.0 - ratio_squared;
    let square_root = radicand.sqrt();
    let negative_l_square_root = -l * square_root;

    let output : f64 = negative_l_square_root - l_cos_expression;

    // let output : f64 = -0.875; // Desmos Line 14
    return output;
}
```

4. Further Research

Polar coordinates allow us to add a sinusoidal wave onto the circular invariant. This gives us the ability to construct multimodal liquidity fingerprints with the following equation where $\alpha = [4, \dots \infty)$ in intervals of 2 and $\beta = \alpha^2$

$$r(\theta) = \frac{L}{\sqrt[\beta]{1 - \frac{1}{2} \sin(\alpha\theta)^2}}. \quad (10)$$

At $\alpha = 4$ our trading function resembles Curve [2]. At $\alpha = 6$ we resemble a bimodal liquidity fingerprint the middle part of which can be compared to our ETHSF hackathon submission for convex liquidity - liquidity that grows as we move away from the current price [13]. We would like to particularly draw attention to $\alpha = 8$ where we have a trimodal liquidity fingerprint which can be very useful for collateralized debt position (CDP) stablecoins such as DAI. We can see empirically that such stablecoins exhibit sinusoidal peaks at 0, 1%, and -1% when minting and burning are activated [Figure4]. We provide an interactive model for further research for such CDP stablecoins in [desmos](#) [14]

5. Disclaimer

This paper is for general information purposes only. It does not constitute investment advice or a recommendation or solicitation to buy or sell any investment and should not be used in the evaluation of the merits of making any investment decision. It should not be relied upon for accounting, legal or tax advice or investment recommendations.

REFERENCES

- [1] Robinson, D: Uniswap v3 - The Universal AMM, <https://www.paradigm.xyz/2021/06/uniswap-v3-the-universal-amm>, last accessed 2025/09/19
- [2] Egorov, M: StableSwap - efficient mechanism for Stablecoin liquidity, https://docs.curve.finance/assets/pdf/whitepaper_stableswap.pdf, last accessed 2025/09/19
- [3] Euler Labs: EulerSwap White Paper, https://github.com/euler-xyz/euler-swap/blob/master/docs/whitepaper/EulerSwap_White_Paper.pdf, last accessed 2025/09/19
- [4] Adams, H et al: Uniswap v3 Whitepaper, <https://uniswap.org/whitepaper-v3.pdf>, last accessed 2025/09/19
- [5] White, D., Robinson, D., Moallemi, C.: Orbital, <https://www.paradigm.xyz/2025/06/orbital>, last accessed 2025/09/19
- [6] Tolstikov, V.: Negative Price AMM, <https://ethresear.ch/t/negative-price-amm/17833>, last accessed 2025/09/19
- [7] Schiarizzi, J. Tolstikov, V.: Orbswap, <https://github.com/cup0Joseph/orbswap>, last accessed 2025/09/19
- [8] Tolstikov, V.: Desmos - Swapping Along Circle in Polar Coordinates, <https://www.desmos.com/calculator/hfccwalpqg>, last accessed 2025/09/19
- [9] Tolstikov, V.: Desmos - Elliptical Liquidity Shift, <https://www.desmos.com/calculator/amk99uyxo3>, last accessed 2025/09/19
- [10] University of Washington School of Law: The Silicon Valley Bank Collapse <https://www.law.uw.edu/news-events/news/2023/svb-collapse>, last accessed 2025/09/19
- [11] Wikipedia: Kessler Syndrome, https://en.wikipedia.org/wiki/Kessler_syndrome, last accessed 2025/09/19
- [12] Tolstikov, V.: Desmos - CCMM and CSEMM, <https://www.desmos.com/calculator/wy74cucie0>, last accessed 2025/09/19
- [13] Wentz, M., Tolstikov, V.: Curvy - Custom Curve, <https://ethglobal.com/showcase/curvy-vjk7e>, last accessed 2025/09/19
- [14] Tolstikov, V.: Desmos - Polarswap, <https://www.desmos.com/calculator/dq1ao5mryr>, last accessed 2025/09/19

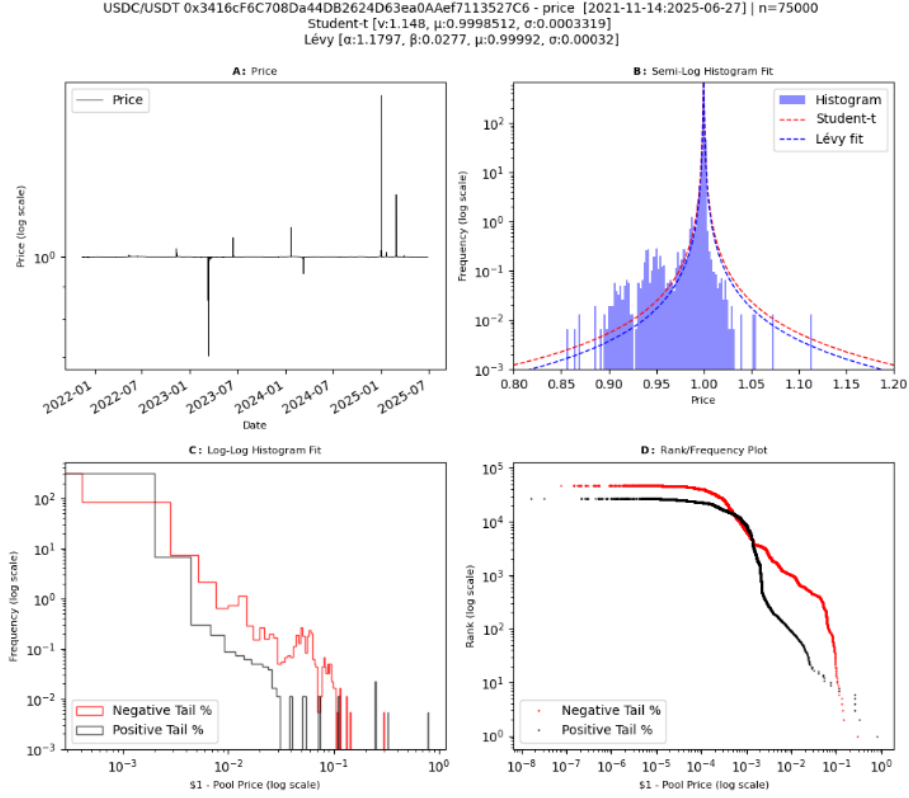


Fig. 1: 1A: Price behavior of USDC/USDT on Uniswap v3 since inception. 1B: Notice the asymmetric skew of the price movement, indicating that providing liquidity to the left of the price makes more sense rather than the right. A personal discussion with Ciamac Moallemi led to the conclusion that the reason for this skew is the redemption fee barrier imposed on one of the stablecoin issuers, which created an incentive to swap the stablecoin on Uniswap into another token without the redemption fee. 1C and 1D in log-log hint at fat-tailed behavior of stablecoin pools. Data retrieved from Dune Analytics from the period since the pool's inception on Uniswap v3 until 2025/06/27 and processed in Python.

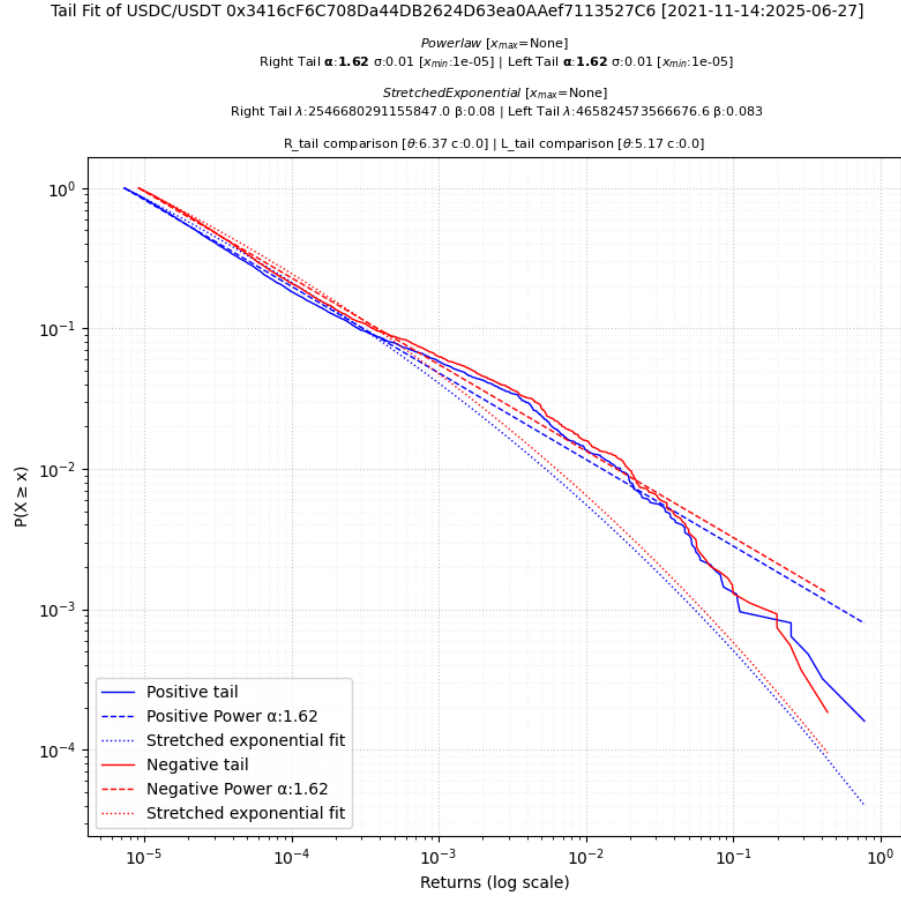


Fig. 2: We have examined the top three most populated and oldest stablecoin pools on Uniswap v3 since inception and found the same pattern for all of them. We have a fat-tail with the power law statistically preferred over the stretched exponential. Its alpha exceeds that of the Cauchy distribution ($\alpha < 2$), making the act of stacking multiple such assets into one pool grow in fragility.

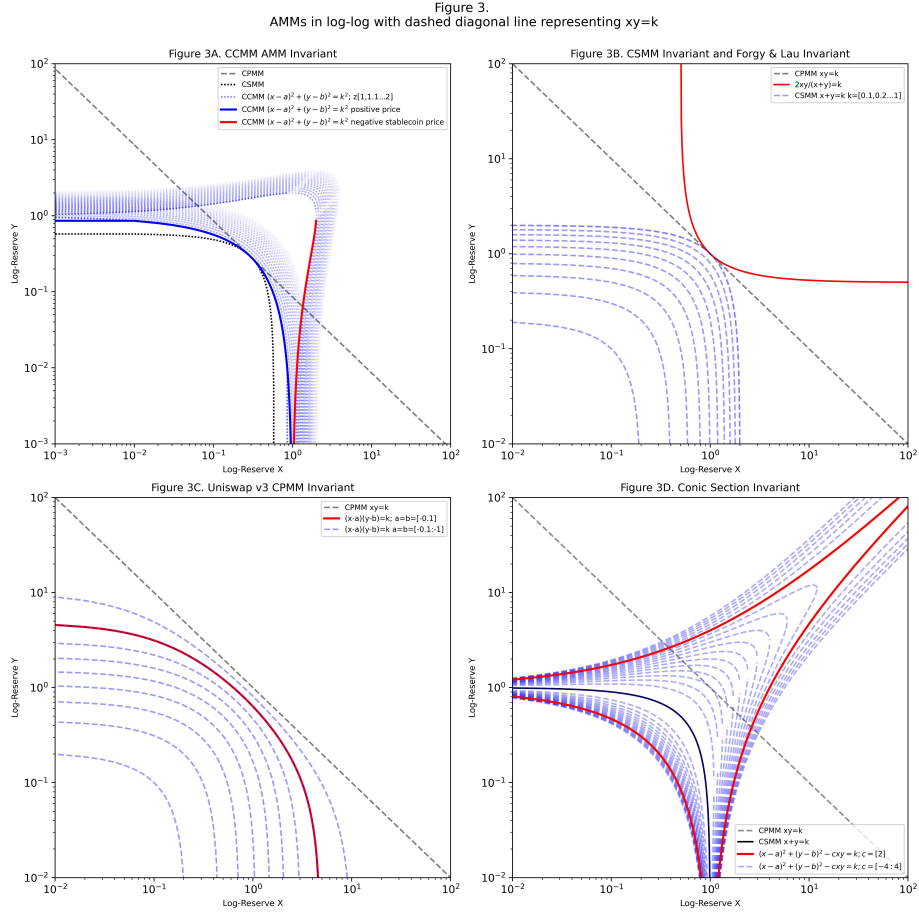


Fig. 3: 3A: CCMM invariant with red line representing a negative price for an asset priced in a currency with a zero bound such as a stablecoin. 3B: Red invariant curve for limiting impermanent loss as outlined by Eric Forgy and Leo Lau with a CSMM in dashed blue for comparison. 3C: Concentrated liquidity in Uniswap v3. 3D: Concentration of liquidity in a negatively priced AMM resembles a Uniswap v3 invariant as long as price does not enter negative territory.

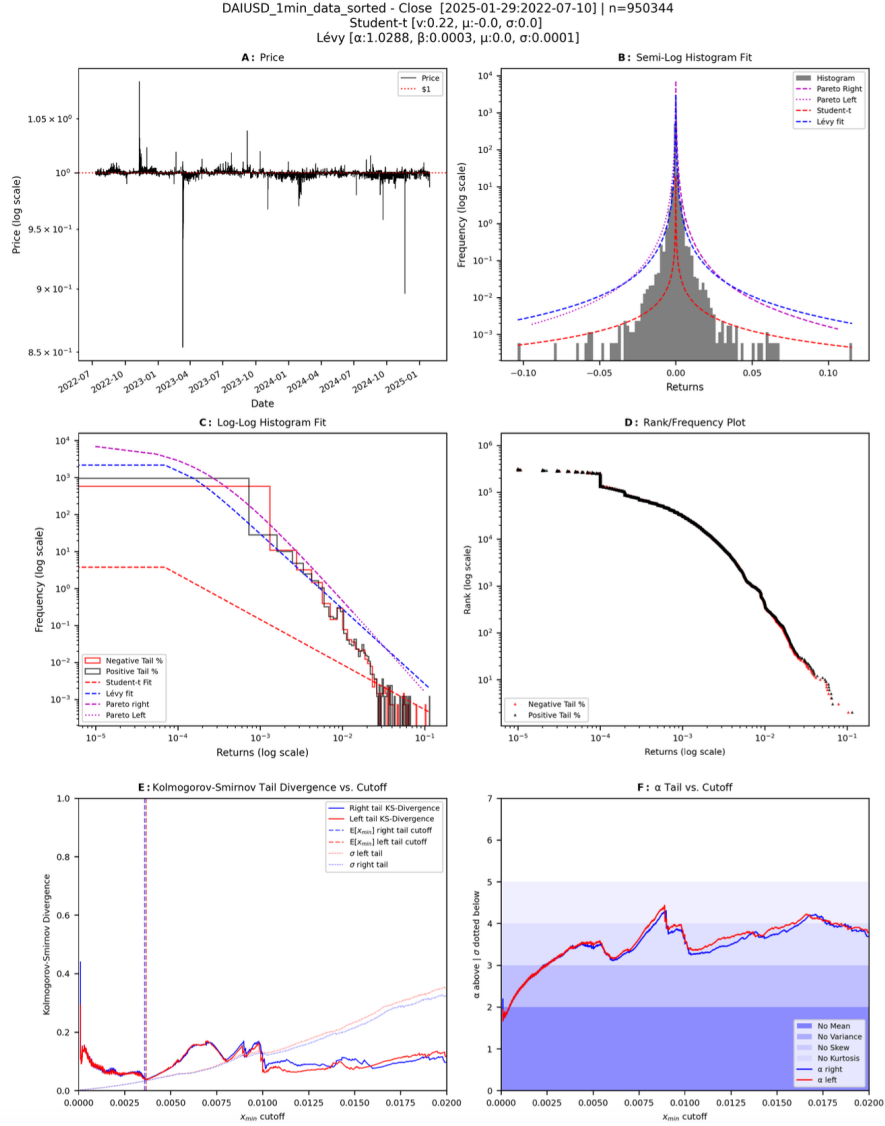


Fig. 4: 4A: DAI-USD Log-Y price dynamics. 4B: Log-Y histogram and fits of Levy, Student-T and Pareto. For stablecoins the Pareto distribution provides the better fit and the Student-T, while focusing on outliers, fails to capture the body giving an inaccurate read. 4C: Log-Log histogram of the tails. Note the histogram bump at 1% has a unique sinusoidal bump. 4D: The rank-frequency plot allows us to look more closely at the outliers. There is a unique step function at the low returns range and a sinusoidal ripple in the tail starting at 1%. 4E: We select the earliest minimum x_{min} cutoff value of the KS-Divergence visualized as dashed verticals in order to minimize our error rate σ . 4F: We map the power law spectrum showing the fatness of the tail and see the sinusoidal bump prior to $x_{min} = 1\%$. Data retrieved from polygon.io for Centralized Exchange Data for the price of DAI from the period 2022/07/10 until 2025/01/29 every one minute time interval and processed in Python using the **powerlaw** library.